

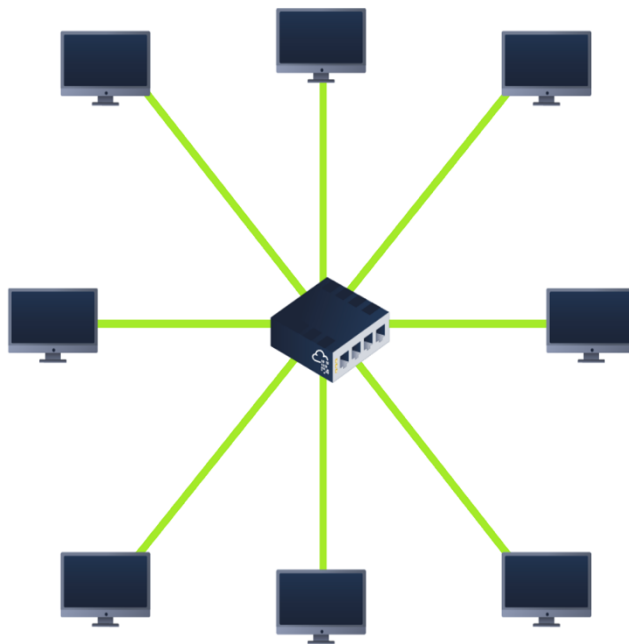
Networking Basics

Introducing LAN Topologies

- **Local Area Network (LAN) Topologies**

Topology is implementation of various network design.

Star Topology~Devices are individually connected via a central networking device e.g switch or hub.



Advantages:

- Reliable
- Scalable

Disadvantages:

- Maintenance is costly
- Troubleshooting faults is much harder
- Prone to failure

BUS Topology

It relies upon single connection know as **Backbone Cable**

All data destined for each device travel along the cable, prone to becoming slow if devices within the topology are simultaneously requesting data



Advantages:

-Easy and cost-efficient to set up

Disadvantages:

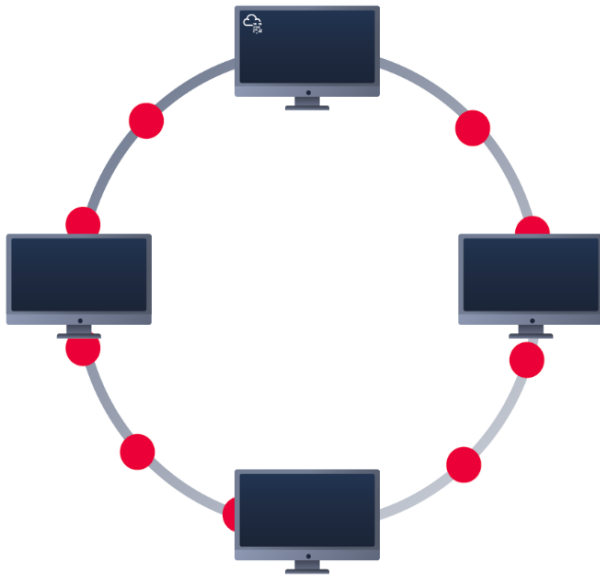
-There is little redundancy in place in case of failure.

Ring Topology (Token Topology)

Devices are connected directly to each other to form a loop.

It send data across the loop until it reaches the destined devices.

Device will only send received data from other devices in this topology



Advantages:

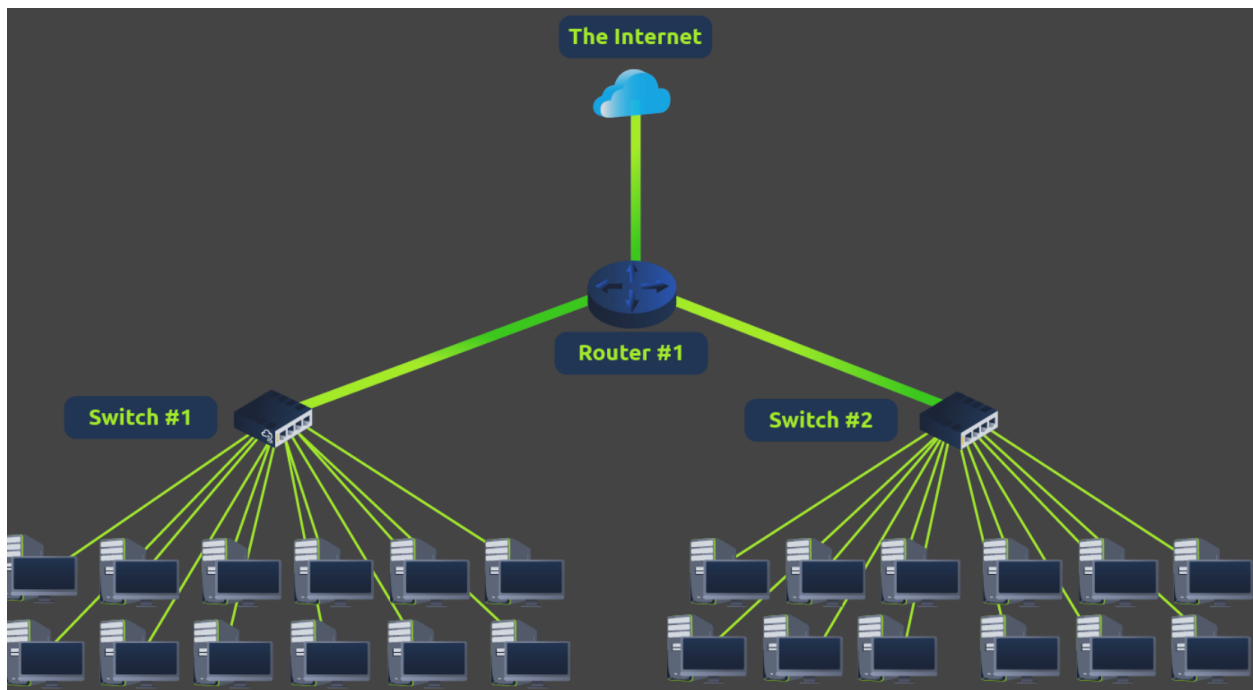
- ~Easy to troubleshoot any faults that arise
- ~less prone to bottleneck

Switch

Its dedicated devices within a network that are designed to aggregate multiple other devices e.g. computers, printers etc

Switch keep track of what device is connected to which port.

This way, when they receive a packet, instead of repeating that packet to every port like a hub would do, it just sends it to the intended target, thus reducing network traffic.



Router

Routers connect networks and pass data between them~its does this by using routing.

Routing involves creating a path between networks so that this data can be successfully delivered

